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use_tokens.py

```
## Assignment 1, Page 6: Use Tokens

from nltk.tokenize import word_tokenize


lorenzo_paragraph = "Lorenzo di Piero de'Medici was an Italian statesman, banker, de facto ruler of the Fl
#print(f'Tokenized Text by Word Using word_tokenize(): {word_tokenize(lorenzo_paragraph)}')
#print('\nTokenized Text by Word Using words(): ')
#print(lorenzo_paragraph.words())
from nltk.tokenize import sent_tokenize


lorenzo_paragraph = "Lorenzo di Piero de'Medici was an Italian statesman, banker, de facto ruler of the Fl
#print(f'Tokenized Text by Sentence Using sent_tokenize(): {sent_tokenize(lorenzo_paragraph)}')
for sent in sent_tokenize(lorenzo_paragraph):
  print(sent)


```

Guide

Collapse Setting Up Your Environment

# Tokenize Text

## Definition: Token

- The process of tokenization involves dividing plain text, such as a phrase, sentence, paragraph, or entire documents into smaller chunks of data so that it is easier to analyze.
- These smaller chunks of data are referred to as tokens, which may include words, phrases, or sentences.

There are two methods commonly used to tokenize text: `word_tokenize` and `sent_tokenize`.

**Tokenizing by word** - The benefit of using tokenization by word is that we can pinpoint words that are frequently used. If we were analyzing a group of restaurant ads in NY, and found that the word "vegan" was used often, then we might assume that there are plenty of vegan options at these restaurants.

```
word_tokenize
```

**Tokenizing by sentence** - When tokenizing by sentence, we have the ability to analyze how the words in the sentence correlate with each other, which allows us to better understand the context of the words. If we were analyzing a group of restaurant ads in NY, and found that the sentence "No vegan options." was used, then we can determine that there are not plenty of vegan options at these restaurants.

```
sent_tokenize
```

## Word Tokens

Using the `nltk.tokenize` package, we can import the `word_tokenize` method from the NLTK library.

For this first exercise, we will work with a string of data assigned to the variable called `lorenzo_paragraph`, as shown in the code below.

```
 with the mature phase of the Italian Renaissance and the Golden Age of Florence."
```

The imported `word_tokenize` method allows us to break this text down by word and punctuation, where each word is grouped in its own string within a list.

Enter the code below, then select the `TRY IT` button to run the file.

```
print(f'Tokenized Text by Word Using word_tokenize(): {word_tokenize(lorenzo_paragr
```

TRY IT

```
Tokenized Text by Word Using word_tokenize():
['Lorenzo', 'di', 'Piero', "de'Medici", 'was', 'an',
'Italian', 'statesman', ',', 'banker', ',', 'de',
'facto', 'ruler', 'of', 'the', 'Florentine',
'Republic', 'and', 'the', 'most', 'powerful', 'and',
'enthusiastic', 'patron', 'of', 'Renaissance',
'culture', 'in', 'Italy', '.', 'Also', 'known', 'as',
'Lorenzo', 'the', 'Magnificent', '(', 'Lorenzo',
'il', 'Magnifico', ')', 'by', 'contemporary',
'Florentines', ',', 'he', 'was', 'a', 'magnate', ',',
'diplomat', ',', 'politician', 'and', 'patron', 'of',
'scholars', ',', 'artists', ',', 'and', 'poets', '.',
'As', 'a', 'patron', ',', 'he', "'s", 'best',
'known', 'for', 'his', 'sponsorship', 'of',
'artists', 'such', 'as', 'Botticelli', 'and',
'Michelangelo', '.', 'He', 'held', 'the', 'balance',
'of', 'power', 'within', 'the', 'Italic', 'League',
',', 'an', 'alliance', 'of', 'states', 'that',
'stabilized', 'political', 'conditions', 'on', 'the',
'Italian', 'peninsula', 'for', 'decades', ',', 'and',
'his', 'life', 'coincided', 'with', 'the', 'mature',
'phase', 'of', 'the', 'Italian', 'Renaissance',
'and', 'the', 'Golden', 'Age', 'of', 'Florence', '.']
```

Your output should display as shown below:

```
Tokenized Text by Word Using word_tokenize(): ['Lorenzo', 'di', 'Piero', "de'Medici
```

▶ **Did you notice?**

**Try this variation:**

Let's use the reader function called `words()` previously used in the earlier pages of this assignment to tokenize the variable called `lorenzo_paragraph` by word again.

Predict what you think will happen, enter the below code, then select the `TRY IT` button to run the file.

```
print('\nTokenized Text by Word Using words(): ')
print(lorenzo_paragraph.words())
```

TRY IT

```
Traceback (most recent call last):
  File "exercises/use_tokens.py", line
8, in <module>
    print(lorenzo_paragraph.words())
AttributeError: 'str' object has no
attribute 'words'
```

Your output should display as shown below.

```
Tokenized Text by Word Using words():
Traceback (most recent call last):
print(lorenzo_paragraph.words())
AttributeError: 'str' object has no attribute 'words'
```

▼ **What does the error mean?**

```
Tokenized Text by Word Using words():
Traceback (most recent call last):
print(lorenzo_paragraph.words())
AttributeError: 'str' object has no attribute 'words'
```

The reader function called `words` is only designed to be used on corpora.

When we need to break non-corpus text data down by word, we tokenize using the `word_tokenize` method.

**Important**: Before moving on, make sure to comment out or delete the following lines from your code.

```
print(lorenzo_paragraph.words())
```

Otherwise, the computer will get stuck on this error when running the remaining exercises on this page.

## Sentence Tokens

Using the `nltk.tokenize` package, we can import the `sent_tokenize` method from the NLTK library.

We will continue to work with the string of data assigned to the variable `lorenzo_paragraph`, as shown in the code below.

```
 with the mature phase of the Italian Renaissance and the Golden Age of Florence."
```

The imported `sent_tokenize` method allows us to break this text down by sentence, where each sentence is grouped in its own string within a list.

Enter the code below, then select the `TRY IT` button to run the file.

```
print(f'Tokenized Text by Sentence Using sent_tokenize(): {sent_tokenize(lorenzo_pa
```

TRY IT

```
Tokenized Text by Sentence Using sent_tokenize():
["Lorenzo di Piero de'Medici was an Italian
statesman, banker, de facto ruler of the Florentine
Republic and the most powerful and enthusiastic
patron of Renaissance culture in Italy.", 'Also known
as Lorenzo the Magnificent (Lorenzo il Magnifico) by
contemporary Florentines, he was a magnate, diplomat,
politician and patron of scholars, artists, and
poets.', "As a patron, he's best known for his
sponsorship of artists such as Botticelli and
Michelangelo.", 'He held the balance of power within
the Italic League, an alliance of states that
stabilized political conditions on the Italian
peninsula for decades, and his life coincided with
the mature phase of the Italian Renaissance and the
Golden Age of Florence.']
```

Your output should display as shown below:

```
Tokenized Text by Sentence Using sent_tokenize(): ["Lorenzo di Piero de'Medici was
```

**Try this variation:**

Write the code to produce the following expected output, then select the `TRY IT` button to run the file and check your work.

The expected output is shown below.

```
Tokenized Text by Sentence, Separating Each Sentence by Line:
- Lorenzo di Piero de'Medici was an Italian statesman, banker, de f
- Also known as Lorenzo the Magnificent (Lorenzo il Magnifico) by c
- As a patron, he's best known for his sponsorship of artists such
- He held the balance of power within the Italic League, an allianc
```

TRY IT

```
Lorenzo di Piero de'Medici was an
Italian statesman, banker, de facto
ruler of the Florentine Republic and the
most powerful and enthusiastic patron of
Renaissance culture in Italy.
Also known as Lorenzo the Magnificent
(Lorenzo il Magnifico) by contemporary
Florentines, he was a magnate, diplomat,
politician and patron of scholars,
artists, and poets.
As a patron, he's best known for his
sponsorship of artists such as
Botticelli and Michelangelo.
He held the balance of power within the
Italic League, an alliance of states
that stabilized political conditions on
the Italian peninsula for decades, and
his life coincided with the mature phase
of the Italian Renaissance and the
Golden Age of Florence.
```

▼ **Solution**

One possible solution is shown below, where we can use a `for` loop along with `sent_tokenize` to tokenize the text by sentence, separating each sentence by line.

```
print('\nTokenized Text by Sentence, Separating Each Sentence by Li
for element in sent_tokenize(lorenzo_paragraph):
  print(f'- {element}')
```

**Reading Question**

**Fill in the blanks.**

We can import and use the word_tokenize method to tokenize text by word.

We can import and use the sent_tokenize method to tokenize text by sentence.

We can import and use the **word_tokenize** method to tokenize text by word.

We can import and use the **sent_tokenize** method to tokenize text by sentence.

Check Will Safe

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